

Project Title

Realistic_E_Commerce_Sales_Analysis

Domain;

E-commerce

Objective

1. Analyze overall sales performance to understand total revenue, number of orders, and sales trends over time.
2. Identify top-selling products and categories to determine which products contribute most to revenue.
3. Analyze customer purchasing behavior such as average order value, purchase frequency, and popular purchase periods.
4. Evaluate regional or location-based sales performance to identify high-performing markets.
5. Study the impact of discounts and promotions on sales and revenue.

Outcome

The project provided valuable insights into e-commerce sales performance, customer purchasing behavior, and product demand. The analysis helped identify top-selling products, seasonal sales trends, and key revenue drivers, enabling businesses to make informed decisions to improve sales strategies and operational efficiency.

Dataset Information

Source: Kaggle

Year / Timeline: Data collected during 2020/2025

Dataset Description:

Name - It tells us about the names of E-commerce customers

Type of Analysis

This project mainly uses Descriptive Analysis and Diagnostic Analysis to understand sales performance, identify trends, and determine factors affecting e-commerce sales

Problem Definition and Dataset Selection

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('/content/realistic_e_commerce_sales_data (1).csv')
```

```
df.head()
```

	Customer ID	Gender	Region	Age	Product Name	Category	Unit Price	Quantity	Total Price	Shipping Fee	Shipping Status	Order Date
0	CUST0268	Male	North	NaN	Monitor	Electronics	300.0	5	1500	13.31	Returned	12/8/2023
1	CUST0046	Male	West	22.0	Headphones	Accessories	100.0	2	200	6.93	In Transit	4/9/2023
2	CUST0169	Female	South	54.0	Monitor	Electronics	300.0	1	300	11.31	Returned	8/28/2023
3	CUST0002	Male	North	23.0	Headphones	Accessories	100.0	5	500	12.22	Delivered	1/18/2023
4	CUST0173	Female	South	NaN	Laptop	Electronics	1500.0	3	4500	5.40	Delivered	1/19/2023

df.tail()

	Customer ID	Gender	Region	Age	Product Name	Category	Unit Price	Quantity	Total Price	Shipping Fee	Shipping Status	Order Date
995	CUST0201	Female	South	49.0	Headphones	Accessories	100.0	1	100	17.21	In Transit	1/5/2023
996	CUST0133	Male	East	47.0	Laptop	Electronics	1500.0	1	1500	19.19	Delivered	4/1/2023
997	CUST0055	Female	North	NaN	Mouse	Accessories	30.0	5	150	19.35	Delivered	10/20/2023
998	CUST0023	Female	South	29.0	Laptop	Electronics	1500.0	5	7500	10.36	Returned	1/7/2023
999	CUST0040	Male	NaN	29.0	Smartwatch	Wearables	200.0	1	200	7.53	Delivered	1/30/2023

df.shape

(1000, 12)

df.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 1000 entries, 0 to 999

Data columns (total 12 columns):

Column Non-Null Count Dtype

--- ---- -

0 Customer ID 1000 non-null object

1 Gender 1000 non-null object

2 Region 950 non-null object

3 Age 900 non-null float64

4 Product Name 1000 non-null object

5 Category 1000 non-null object

6 Unit Price 1000 non-null float64

7 Quantity 1000 non-null int64
 8 Total Price 1000 non-null int64
 9 Shipping Fee 1000 non-null float64
 10 Shipping Status 950 non-null object
 11 Order Date 1000 non-null object

dtypes: float64(3), int64(2), object(7)

memory usage: 93.9+ KB

df.describe()

	Age	Unit Price	Quantity	Total Price	Shipping Fee
count	900.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	46.695556	457.703777	3.008000	1346.600000	12.416390
std	15.011400	537.231434	1.404246	1834.037877	4.412185
min	18.000000	30.000000	1.000000	30.000000	5.000000
25%	35.000000	50.000000	2.000000	200.000000	8.560000
50%	49.000000	200.000000	3.000000	600.000000	12.315000
75%	59.000000	800.000000	4.000000	1500.000000	16.075000
max	69.000000	3109.560612	5.000000	7500.000000	19.980000

df.isnull().sum()

	0
Customer ID	0
Gender	0

	0
Region	50
Age	100
Product Name	0
Category	0
Unit Price	0
Quantity	0
Total Price	0
Shipping Fee	0
Shipping Status	50
Order Date	0

dtype: int64

df.duplicated().sum()

np.int64(0)

Stage 2 – Data Cleaning and Pre-processing

df['Age'] = df['Age'].fillna(df['Age'].mean())

df['Shipping Status'] = df['Shipping Status'].fillna(df['Shipping Status'].mode()[0])

df['Region'] = df['Region'].fillna(df['Region'].mode()[0])

```
df.isnull().sum()
```

	0
Customer ID	0
Gender	0
Region	0
Age	0
Product Name	0
Category	0
Unit Price	0
Quantity	0
Total Price	0
Shipping Fee	0
Shipping Status	0
Order Date	0

```
dtype: int64
```

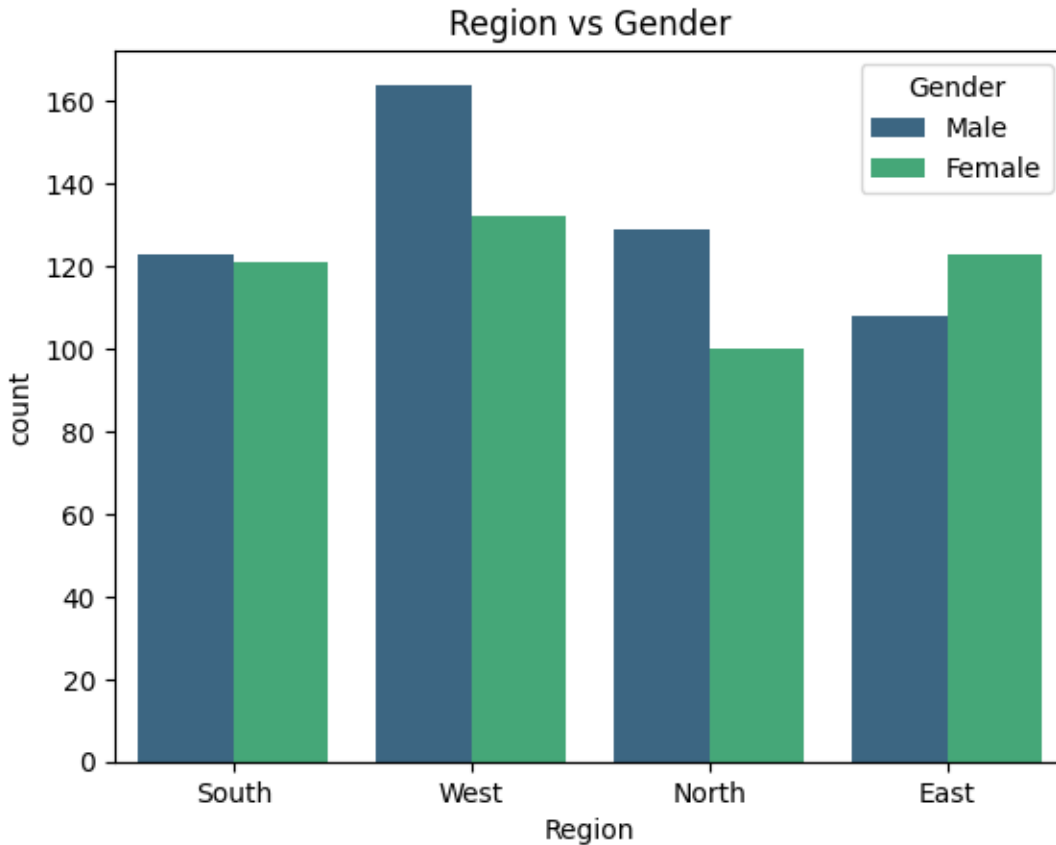
```
df.to_csv('/content/cleaned_dataset.csv',index=False)
```

Stage 3 – EDA and Visualizations

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

Counter plot:

```
sns.countplot(
    x='Region',
    hue='Gender',
    data=df,
    palette='viridis',
    order=['South', 'West', 'North', 'East']
)
plt.title('Region vs Gender')
plt.show()
```



The count plot illustrates the distribution of individuals across four regions (South, West, North, and East) categorized by gender. The visualization helps compare male and female representation in each region. From the plot, we can observe differences in gender distribution across regions and identify which region has the highest participation.

BAR PLOT:

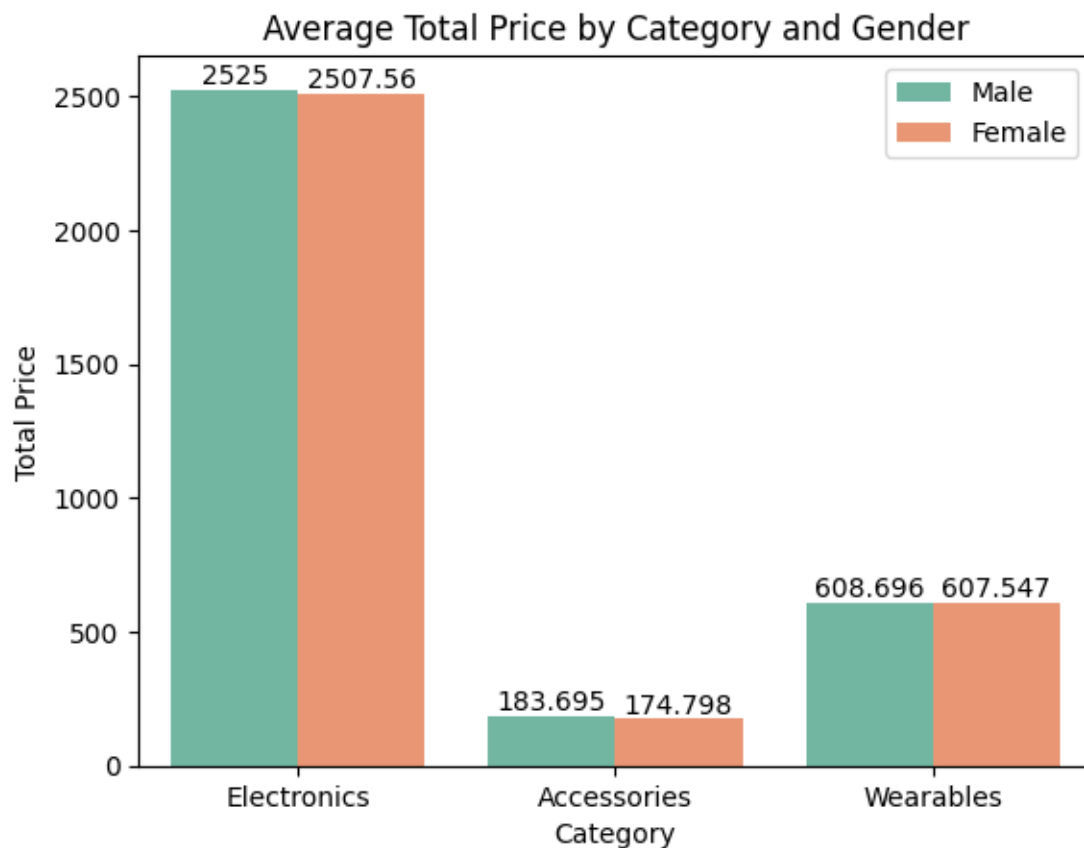
```
bar=sns.barplot(  
    x='Category',  
    y='Total Price',  
    hue='Gender',  
    data=df,  
    estimator='mean',  
    palette='Set2',
```

```

        errorbar=None
    )

plt.title('Average Total Price by Category and Gender')
plt.legend(bbox_to_anchor=(1,1))
plt.bar_label(bar.containers[0])
plt.bar_label(bar.containers[1])
plt.show()

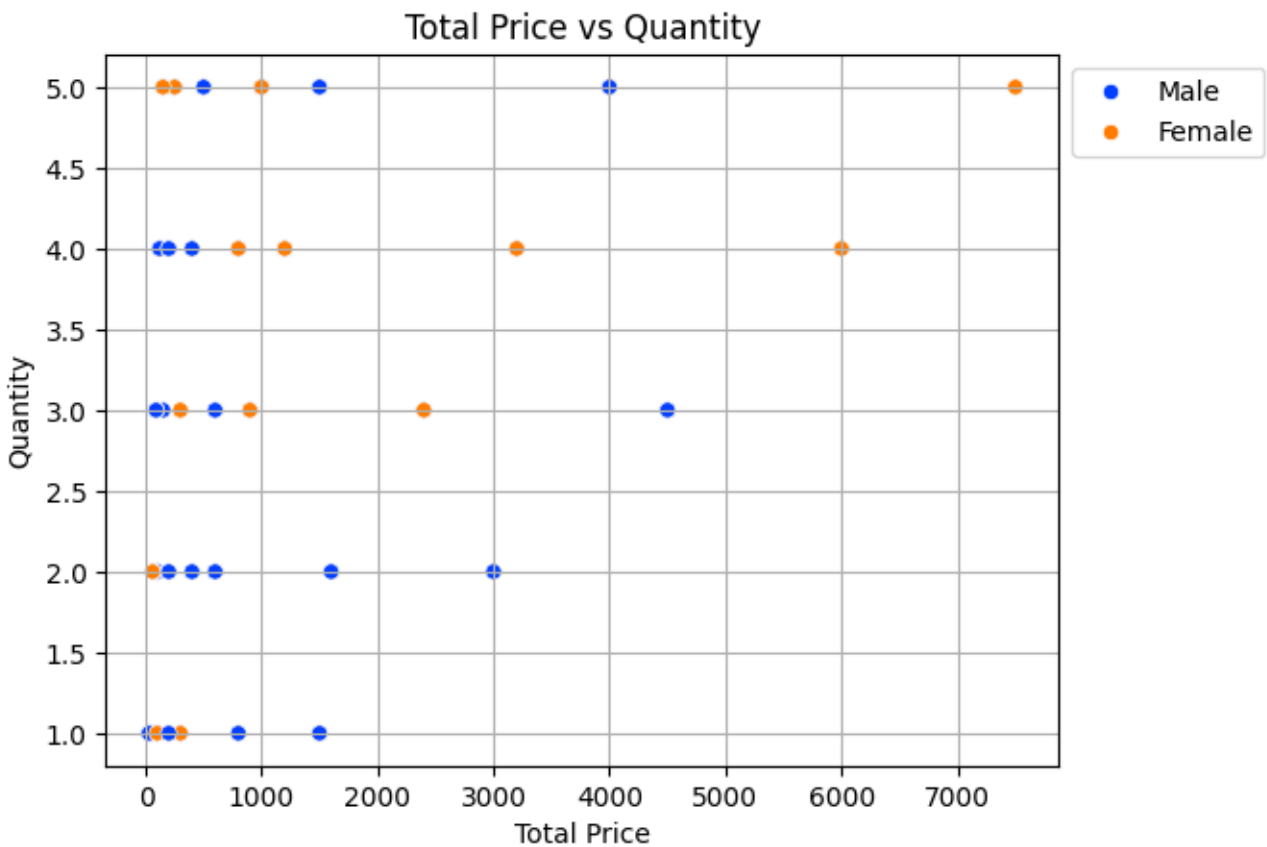
```



The bar plot displays the average total price for different product categories segmented by gender. It allows comparison of purchasing behavior between male and female customers across categories. From the visualization, we can identify which category has the highest average spending and observe gender-based differences in purchasing patterns.

scatter plot:

```
sns.scatterplot(  
    data=df,  
    x='Total Price',  
    y='Quantity',  
    hue='Gender',  
    palette='bright'  
)  
plt.legend(bbox_to_anchor=(1,1))  
plt.title('Total Price vs Quantity')  
plt.grid(True)  
plt.show()
```



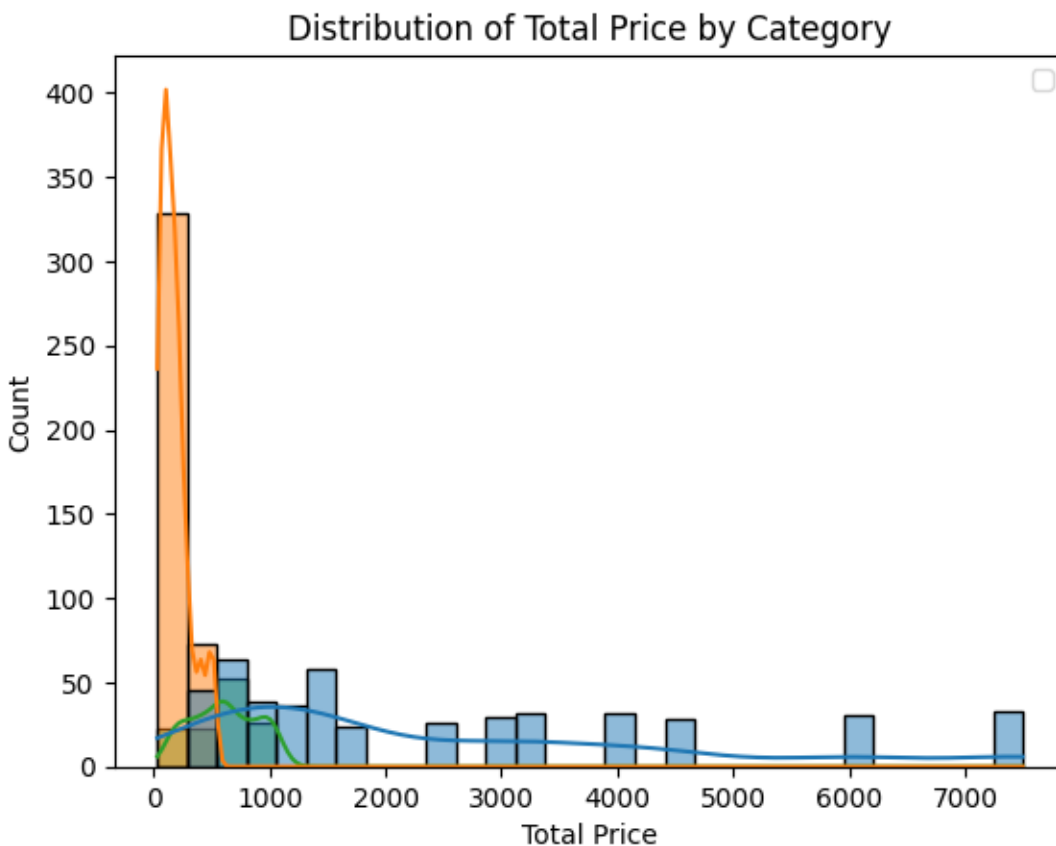
The scatter plot visualizes the relationship between total price and quantity purchased. The plot indicates a positive relationship, where higher quantities generally lead to higher total prices. Additionally, the color grouping by gender helps compare purchasing patterns between male and female customers.

Histogram

```
sns.histplot(data=df, x='Total Price', hue='Category', kde=True)
plt.title('Distribution of Total Price by Category')
plt.legend(bbox_to_anchor=(1,1))
plt.xlabel('Total Price')
plt.ylabel('Count')
plt.show()
```

/tmp/ipykernel_10466/2630544337.py:3: UserWarning: No artists with labels found to put in legend. Note that artists whose label start with an underscore are ignored when legend() is called with no argument.

```
plt.legend(bbox_to_anchor=(1,1))
```



The histogram visualizes the distribution of total purchase prices across different product categories. The plot shows how frequently transactions occur within specific price ranges. By separating the distribution by category, we can compare pricing patterns across product groups. The KDE curve highlights the overall trend and helps identify the most common purchase price ranges.

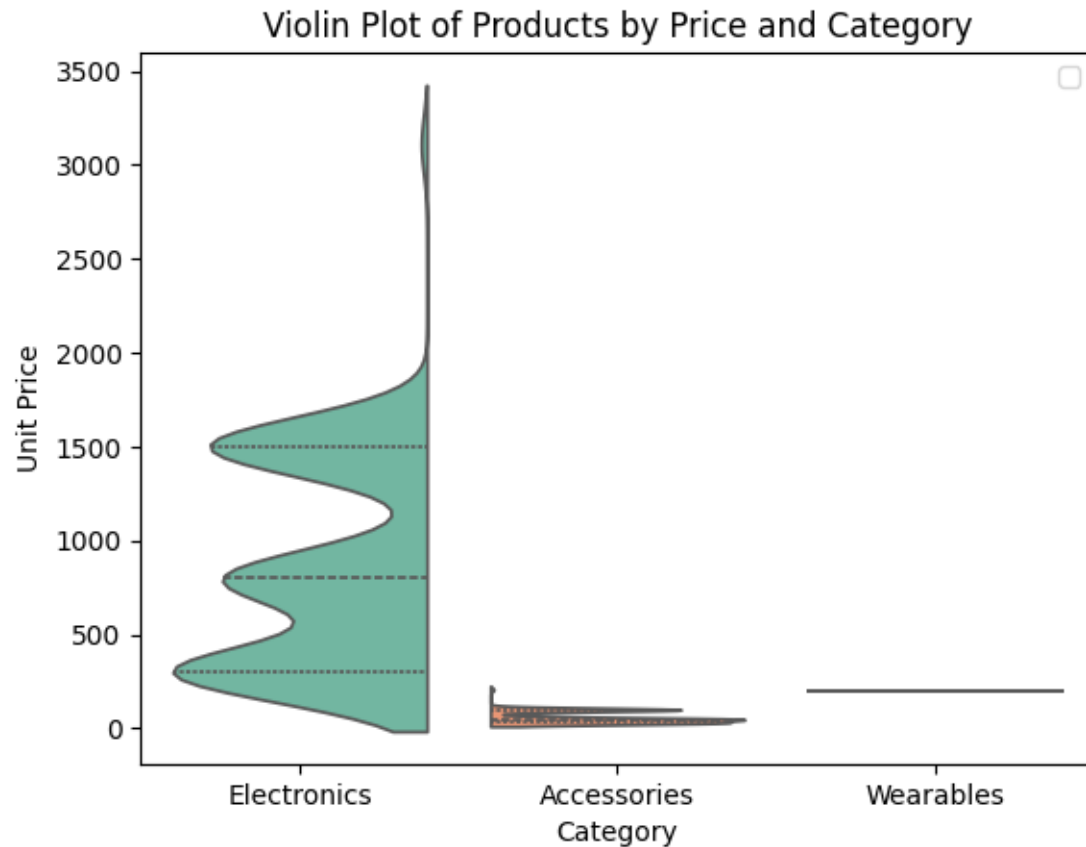
Violinplot:

```
sns.violinplot(  
    x="Category",  
    y="Unit Price",  
    data=df,  
    split=True,          # split by hue -- True  
    inner="quartile",    # box, quartile, point, stick  
    palette="Set2"  
)  
plt.title("Violin Plot of Products by Price and Category")  
plt.legend(bbox_to_anchor=(1,1))  
plt.show()
```

/tmp/ipykernel_10466/912956908.py:1: FutureWarning:

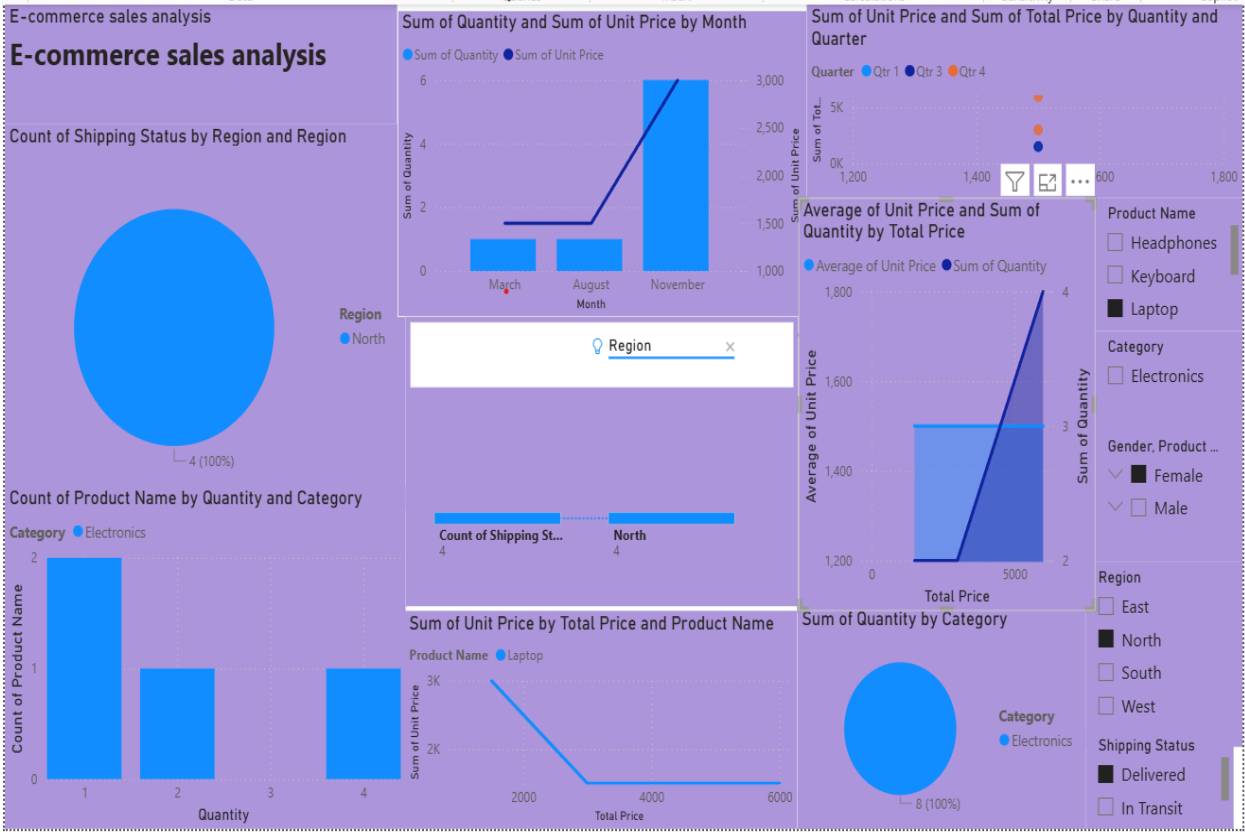
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.violinplot(  
/tmp/ipykernel_10466/912956908.py:10: UserWarning: No artists with labels  
found to put in legend. Note that artists whose label start with an  
underscore are ignored when legend() is called with no argument.  
    plt.legend(bbox_to_anchor=(1,1))
```



The violin plot illustrates the distribution of unit prices across different product categories. The width of each violin represents the density of product prices within that category, while the quartile lines highlight the central price distribution. This visualization helps identify categories with higher price ranges and greater price variability.

DASHBOARD



Conclusion

The E-Commerce Sales Analysis project provides valuable insights into customer purchasing behavior, product performance, and overall sales trends. By analyzing the dataset using Python, Pandas, Matplotlib, and Seaborn, meaningful patterns and relationships were identified.

The analysis revealed which product categories generate the highest revenue and which products are purchased most frequently. Visualizations such as histograms, scatter plots, and violin plots helped understand the distribution of prices, quantities purchased, and customer purchasing patterns across different categories and regions.

Additionally, the study highlighted the relationship between total price and quantity purchased, helping to understand customer spending behavior. The results can help businesses improve pricing strategies, manage inventory efficiently, and focus marketing efforts on high-performing product categories.

Overall, this project demonstrates how data analytics can support better business decisions in the e-commerce industry by transforming raw sales data into meaningful insights.

